AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

# TM311
# Information Security

## Mid-term Examination MTA - Makeup
## Fall Semester 2025/2026

## Date: 3/Dec. /2025

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| **Number of Exam Pages:** 3 (including this cover sheet) | **Time Allowed:** 2 hours |

## Instructions:

- Total Marks: **60 marks**
- This exam consists of 3 parts.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Calculators** are **Not** allowed.

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## PART 1: True/False Questions [10 Marks]

**Questions 1 [10 marks] (2 marks/each):** (True/False)

1. **The primary objective of system access controls is to maximize system performance.**
2. **The IP address of the computer is an example of personally identifiable information (PII)?**
3. **The purpose of a privacy risk assessment is to determine the system usability.**
4. **A Certificate Authority (CA) in PKI is responsible for encrypting data with a public key.**
5. **Privacy by Design involves integrating privacy from the start of product development**

## PART 2: Short Answer Questions [20 Marks]

**P2-Q1 [5 marks]:** Given the scenario of an active attack attempting to disrupt service availability, how can an organization ensure its system remains accessible?

**P2-Q2 [5 marks]:** What is the significance of Privacy Engineering?

**P2-Q3 [5 marks]:** How would you implement a secure messaging system within an organization using PKI, ensuring the integrity and confidentiality of the messages?

**P2-Q4 [5 marks]:** Why is it important to regularly review and update access permissions?

## PART 3: Problem Solving / Critical Thinking Questions [30 Marks]

**P3-Q1 [10 marks]:** A financial institution has discovered that customer account details have been exposed due to a cybersecurity incident. Outline the steps the institution should take to address this breach, ensuring compliance with relevant data protection laws and regulations.

**P3-Q2 [8 marks]:** Given the following real-world problem and answer the questions below:

"You are an IT consultant hired by a healthcare organization (HO) that is undergoing digital transformation to improve its patient care and operational efficiency. HO plans to implement a new system that integrates electronic health records (EHR), appointment scheduling, and billing. This system will handle a large amount of patient data, including medical histories, treatment plans, and payment information, and will



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be accessed by various employees across different departments, including doctors, nurses, administrative staff, and billing personnel."

P3-Q2-1 [4 marks]: How would you design a system access strategy to ensure that employees in medical, administrative, and billing departments have appropriate access to patient data relevant to their roles?

P3-Q2-2 [4 marks]: What measures would you take to protect sensitive patient data, such as medical histories and payment information, from potential internal and external threats?

**P3-Q3 [4 marks]**: Given the rise of IoT devices, how would you address the increased privacy threats associated with these devices in an organizational setting?

**P3-Q4 [8 marks]:** Given the following real-world problem and answer the questions:

"You are an IT consultant hired by a global retail company that is undergoing digital transformation to improve its online and in-store shopping experience. The company plans to implement a new system that integrates online shopping, inventory management, and customer support. This system will handle a large amount of customer data, including payment information and purchase history, and will be accessed by various employees across different departments, including sales, customer service, and logistics."

P3-Q4-1 [4 marks]: How would you design a system access strategy to ensure that employees in sales, customer service, and logistics have appropriate access to customer data relevant to their roles?

P3-Q4-2 [4 marks]: What measures would you take to protect sensitive customer data, such as payment information, from potential internal and external threats?

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**End of questions**

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